

Applications of Deep Learning algorithms for retinal diseases diagnosis based on Optical Coherence Tomography imaging

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Abstract—The accurate and early detection of retinal diseases is critical for effective treatment and prevention of vision loss. Optical coherence tomography (OCT) imaging has become an essential tool for non-invasive diagnosis of retinal diseases. In this paper, a deep learning approach for the automated diagnosis of retinal diseases using OCT images is presented. Specifically, five different deep learning models were evaluated, including DenseNet121, DenseNet169, DenseNet201, InceptionResNet, and a 12 Convolutional layers-based model, and compare their performance in terms of accuracy, AUC, f1-score, and loss. Our results demonstrate that the DenseNet169 model achieved the highest accuracy (97%), AUC (0.997), f1-score (0.97) and a loss of 0.1 among the models evaluated. These results indicate the potential of deep learning models for the automated diagnosis of retinal diseases based on OCT images, and highlight the importance of continued research and development in this field to improve patient outcomes.

Keywords—deep learning, OCT, retinal diseases, early detection

I. INTRODUCTION

The use of deep learning techniques in the field of medical image analysis has gained significant attention over the past few years. In particular, the detection of retinal diseases based on optical coherence tomography (OCT) images has been an area of focus due to the prevalence and severity of retinal diseases such as age-related macular degeneration, diabetic retinopathy, and glaucoma. These diseases are leading causes of vision impairment and blindness worldwide, and early detection is crucial for effective treatment and prevention of further damage [1]. Traditional approaches for retinal disease detection involve manual interpretation of OCT images by experienced clinicians. However, this method is time-consuming, subjective, and prone to errors, which can lead to misdiagnosis and delayed treatment. On the other hand, deep learning techniques have shown promise in automating the detection of retinal diseases, providing a more accurate and efficient alternative to traditional methods.

Deep learning models are trained on large datasets of OCT images, allowing them to learn complex patterns and features that are indicative of retinal diseases. These models use convolutional neural networks (CNNs) to extract meaningful features from OCT images and classify them into different disease categories. The performance of these models has been shown to surpass that of traditional methods, with high

sensitivity and specificity rates. One of the main advantages of deep learning-based approaches is their ability to detect subtle changes in OCT images that may be missed by human experts. For example, early signs of diabetic retinopathy, such as microaneurysms and hard exudates, can be difficult to detect with the naked eye but can be accurately identified by deep learning models [2]. This early detection can lead to timely treatment and prevention of further damage, potentially saving patients from vision loss. Furthermore, deep learning models can be trained to classify OCT images into different disease stages, allowing for personalized treatment plans and disease monitoring. For instance, the severity of age-related macular degeneration can be graded based on the presence of drusen and geographic atrophy, which can be accurately detected and classified by deep learning models. Despite the promising results of deep learning-based approaches, there are still several challenges that need to be addressed: the lack of diverse and representative datasets, which can lead to biased and inaccurate models. Additionally, the interpretability of deep learning models is still a major concern, as it can be difficult to understand how the models make their decisions.

This paper proposes several deep learning techniques that proved good performances for the early detection of retinal diseases, showing great potential in providing accurate and efficient diagnosis and monitoring of these diseases. However, further research is needed to address the remaining challenges and to ensure that deep learning models are reliable and interpretable in a clinical setting. The paper is organized as follows: an introduction into OCT imaging is given in Section II, several AI-based algorithms for retinal diseases diagnosis from the literature review is provided Section III and the methods and results are presented in Section IV and V, respectively. Some conclusions are drawn in the end. The novelty of the paper is addressed to the importance of OCT scans towards the identification of retinal diseases and to the variety of deep learning models that have been applied on an OCT dataset in order to more accurately detect certain conditions.

II. IMAGING THE RETINA: OPTICAL COHERENCE TOMOGRAPHY

The retina is one of the most important structure of the human body responsible for vision. It converts the light that enters the eye into an electrical impulse that is sent to the brain via the optic nerve.

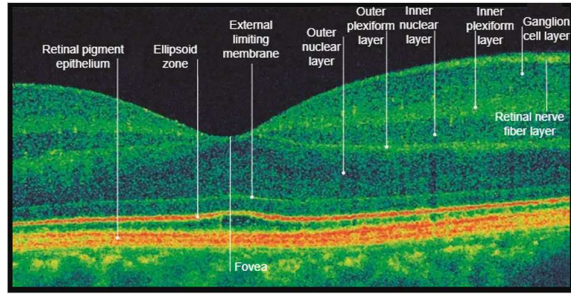


Fig. 1. An OCT scan and the retinal layers (adapted from [5])

The retina is a thin piece of tissue with 0.5 mm thickness which covers the back of the eye. Towards the center of the retina, lies the optic nerve, a white portion that measures 2 x 1.5 mm. From the optic nerve center, the blood vessels of the retina emerge in order to vascularize it. The central section of the retina is represented by the circular field of about 6 mm surrounding the fovea and the entire retina is a circular portion of 30-40 mm in diameter [3].

A cross-sectional view of the retina that provides its distinctive layers can be obtained using the OCT. The OCT is a non-invasive, high-resolution imaging technique that enables the visualization of the anterior eye together with the morphologic features of the retina [4].

The main layers that are visible from an OCT scan are highlighted in Figure 1. From right to left, there is the *retinal nerve fiber layer* (RNFL) which is an extension of the optic nerve fibers and its thickness is of great importance for ocular diseases detection. The *ganglion cells* send impulses to the brain through the optic nerve fibers. The *inner nuclear layer* encompasses horizontal, bipolar and amacrine cells. The *outer plexiform layer* contains the synapses that link the photoreceptors to the bipolar and horizontal cells dendrites. The *outer nuclear layer* includes the cell bodies of the photoreceptors (the rods and cones). The *external limiting membrane* divides the layers of the photoreceptors from the previous layer and is considered as a linear junction between Muller cells which are considered to play an important part in sustaining the structural integrity of the retina [6], and photoreceptors [7]. The *ellipsoid zone* is recognized as the inner/outer segment of the rods and cones layer and it is correlated to the second hyper-reflective band on an OCT scan [8]. The *retinal pigment epithelium* contains the molecules that absorb and filters the light as it passes through the layers and it protects the neural region of the retina. The fovea represents the central part of the retina and it appears as a small depression within the neurosensory retina in which the visual acuity is considered the uppermost [9]. The measurements provided by the OCT are critical in retinal diseases diagnosis, for example in assessing ocular diseases, such as choroidal neovascularization (CNV), diabetic macular edema (DME) or the appearance of yellow deposits, known as DRUSEN which is a condition that may indicate the progress of early-stage dry macular degeneration.

However, considering the fact that there is a huge amount of OCT data that is being generated for evaluating the existence of certain abnormalities, manually assessing the presence of retinal diseases can be subjectively interpreted and may lead to inconsistencies, delaying precise diagnosis and delivering inappropriate treatment. Automatically detecting any retinal condition on OCT images, with the help of

Artificial Intelligence (AI)-based methods and algorithms, would help physicians in more effectively assessing and tackling eye disorders.

III. AI-BASED ALGORITHMS FOR RETINAL DISEASES DIAGNOSIS

AI has the capability to revolutionize early diseases prediction through the integration of recently new developed technologies, such as Machine Learning (ML) and Deep Learning (DL) techniques. This can significantly boost the detection of various conditions and enhance the patient experience through delivering effective treatment. Over the past years, ML and DL models have augmented the state of the art in many medical domains, including ophthalmology as they allow the recognition and quantification of different pathological features related to almost any diseases associated with macula or the retina. The DL-based methods for retinal analysis may vary according to the applicability, understandability or accuracy and depending on the datasets and retinal conditions.

Age-related macular degeneration (AMD) or DME conditions can be assessed from Spectral Domain OCT (SD-OCT) and, using retinal features (retina thickness, retinal layers thickness), can be used for classification and early detection support. ML algorithm have been used for early detection of these diseases [10, 11], with performant outputs. Random Forest and Support Vector Machines obtained the best scores for the classification of the data. DL have showed interesting results in the retinal diseases classification. Tan et al. [12] used fourteen-layer deep Convolutional Neural Network (CNN) for the early detection of AMD with a 91.17% accuracy with blindfold cross-validation and a 95.45% accuracy with a ten-fold cross-validation strategy. Other DL architectures have also provided promising solutions in this direction: VGG-16 [13], with an accuracy of 98.6%, Inception V3 [14], having a 96.6% accuracy, or U-Net and DenseNet [15] and, more recently, Multi-ResNet50 Ensembling [16], scoring a performance of 97.3% accuracy. A study that proposes the classification of CNV, DME, Drusen and Normal OCT images was described in [17]. They adopted several models (VGG16, VGG19, ResNet50, ResNet152, DenseNet121 and InceptionV3), concluding with multiple binary classifier having a performant score: the model using VGG16, VGG19 and InceptionV3 acquired a 98.7% accuracy, 0.987 sensitivity and 0.996 specificity.

IV. METHODS

A. Dataset

A publicly available dataset with 84,484 OCT images, gathered from 4,686 patients, was used for classification [18]. The number of images for training and testing are presented in the following table (Table I).

TABLE I. TRAINING AND TESTING DATASETS

	Class			
	<i>CNV</i>	<i>DME</i>	<i>Drusen</i>	<i>Healthy retina</i>
Train	37,205	11,348	8,616	26,315
Test	250	250	250	250

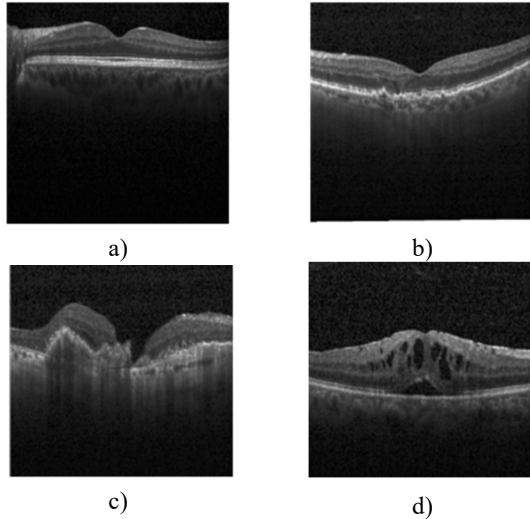


Fig. 2. Representative OCT images from the dataset (a) healthy retina OCT; b) retina affected by Drusen; c) retina affected by CNV; d) retina affected by DME).

Figure 2 depicts the most representative images from the dataset: a healthy retina (a), and other retinal diseases-related OCTs. When it comes to Drusen (b) is characterized by a bumpy RPE layer, rather than the usual flat shape. CNV (c) involves an anomalous increase in the number of vessels from the choroidal vasculature to the neurosensory retina via Bruch's membrane. This may lead to variability in the areas below or above the retinal pigment epithelium (RPE) or within the retina itself, as evidenced by the presence of macular fluid, a bumpy RPE layer, and poorly defined boundaries. Finally, DME (d) is characterized by the accumulation of intraretinal fluids between the inner limiting membrane (ILM) and the RPE layer, which can be visualized as dark-colored areas.

B. Data augmentation

Data augmentation is a technique widely used in deep learning to increase the size of training datasets by artificially generating new samples through various transformations of the original data. The objective of data augmentation is to improve the performance of ML/DL models by reducing overfitting and increasing generalization. Data augmentation is particularly important in medical image analysis, where datasets are often limited due to the difficulty and expense of obtaining and labeling medical images. One popular method of data augmentation in medical image analysis is to use the Keras ImageDataGenerator function, which provides a range of image transformation options such as rotation, scaling, and flipping. ImageDataGenerator was easily integrated into the development process, and the generated images were fed directly into the training process.

C. Proposed models

The proposed models for the classification of retinal diseases were based on: DenseNet, InceptionResNet and a neural network based on 12 convolutional layers. DenseNet and InceptionResNet are two popular deep learning models for image classification tasks, including medical image analysis. DenseNet is a neural network architecture that connects each layer to every other layer in a feed-forward fashion, creating dense connections between layers. This allows for efficient feature reuse and reduces the number of

parameters required, leading to improved accuracy and reduced overfitting. DenseNet has been shown to achieve state-of-the-art performance on a range of image classification tasks, including medical image analysis. On the other hand, Inception-ResNet is a combination of Inception and ResNet models. Inception modules use multiple filters of different sizes to extract features, while ResNet modules use residual connections to improve gradient flow and prevent vanishing gradients. The combination of these two models provides the benefits of both models and results in improved accuracy. The dataset was split on 83,484 images for the training set from all 4 classes and 1,000 for testing and validation, from all 4 classes. The used models were DenseNet121, DenseNet169, DenseNet201, InceptionResNet and InceptionV3. They were all trained on Google Colaboratory [19], on a Tesla T4 GPU, for 10 epochs. The results are presented in the next sub-section.

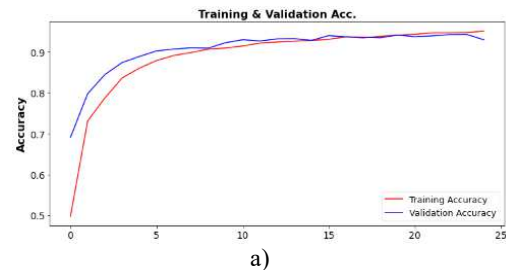
In addition to these models, a convolutional layers-based model was also proposed. It consists of 12 convolutional layers, followed by max pooling layers, dropout layers, and fully connected layers. The activation function used in all convolutional layers is Rectified Linear Unit (ReLU), while the output layer uses Softmax activation to generate the class probabilities. The model is compiled with the Adam optimizer, learning rate of 0.001, and categorical cross-entropy loss function. The dropout layers are used to prevent overfitting, and the number of nodes in the fully connected layers is gradually reduced towards the output layer. The model has a total of 2,320,565 trainable parameters, were adjusted to optimize the model's performance on the given dataset. This model was trained on 25 epochs.

V. EXPERIMENTAL RESULTS AND DISCUSSION

The results for the above-mentioned classifiers are described in the following table (Table II). The table presents the results of evaluating five different models for a classification task. The models evaluated are DenseNet121, DenseNet169, DenseNet201, InceptionResNet, and a 12 Convolutional layers-based model.

TABLE II. RESULTS

Model	Accuracy (%)	AUC	f1-score	Loss
DenseNet121	95.56	0.996	0.94	0.11
DenseNet169	97	0.997	0.97	0.1
DenseNet201	96	0.997	0.96	0.12
InceptionResNet	93.18	0.99	0.93	0.18
12 Convolutional layers-based model	93	0.95	0.93	0.20



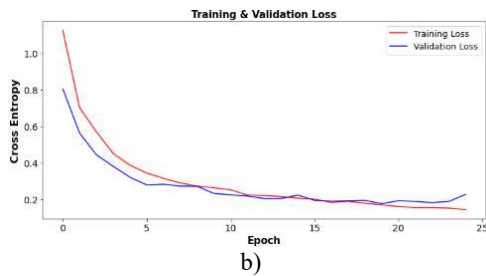


Fig. 3. 12 Convolutional layers-based model: a) Training and Validation accuracy; b) Training and Validation Loss

The evaluation metrics used are accuracy, AUC (Area Under the ROC Curve), f1-score, and loss. Accuracy measures the proportion of correctly classified instances, while AUC measures the ability of the model to distinguish between positive and negative instances. The f1-score is the harmonic mean of precision and recall, and it is a measure of the model's overall performance. Loss is the objective function that the model is optimized to minimize during training. Based on the presented results, DenseNet169 achieves the highest accuracy (97%), AUC (0.997), and f1-score (0.97) and loss value (0.1) among the models evaluated, while DenseNet201 comes second with slightly lower scores. InceptionResNet and the 12 Convolutional layers-based model achieved lower scores, but comparable to the DenseNets. The training-validation accuracy and loss graphs (Fig. 3) were also chosen to be presented, for the model with 12 Convolutional layers as the performances were substantially suitable and comparable to the multiple layers-based architectures.

VI. CONCLUSION

In conclusion, the application of deep learning models for the detection of retinal diseases, such as Drusen, CNV, and DME, using OCT images has shown promising results. The models demonstrated high accuracy, sensitivity, and specificity in detecting and classifying these diseases, which can help clinicians in making accurate and timely diagnoses. The proposed deep learning models for retinal disease detection can potentially lead to earlier detection, better patient outcomes, and reduced healthcare costs. However, some limitations of the model are represented by the similarities with the state-of-the-art results and the challenges that still need to be addressed, such as the need for large and diverse datasets, and the potential for model overfitting. Future research should focus on improving the performance of deep learning models for retinal disease detection by incorporating new imaging modalities and developing more robust models that can handle variations in image quality and other confounding factors. Additionally, efforts should be made to validate the models on larger and more diverse patient populations to ensure their generalizability. Overall, deep learning models have the potential to revolutionize the field of retinal disease detection, and I believe that further development and research in this area can significantly improve patient outcomes and advance the field of ophthalmology.

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